

Saima Afrin

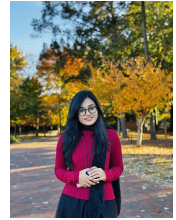
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Education

- 2022 – Present **PhD Candidate** in Computer Science, William & Mary
Research Focus: *Green AI for Software Engineering*
- 2015 – 2019 **B.Sc. Computer Science and Engineering** in Daffodil International University, Bangladesh.

Research Publications (Ph.D.)

- 1 **S. Afrin**, M. Z. Haque, and A. Mastropaolo, “Quantize with confidence? an empirical study of quantization for code generation,” in *2026 IEEE International Conference on Software Maintenance and Evolution (ICSME)*, IEEE, 2026.
- 2 **S. Afrin**, A. Midolo, C. Escobar-Velásquez, M. Linares-Vásquez, W. Ding, B. Xu, M. Di Penta, and A. Mastropaolo, “Large language models for code generation from multilingual prompts: A curated benchmark and a study on code quality,” *Under review at ACM Transactions on Software Engineering and Methodology (TOSEM)*, 2026.
- 3 A. Giagnorio, A. Mastropaolo, **S. Afrin**, M. Di Penta, and G. Bavota, “Evaluating the impact of post-training quantization on large language models for code generation,” in *2026 IEEE/ACM 34th International Conference on Program Comprehension (ICPC)*, IEEE, 2026.
- 4 M. Z. Haque, **S. Afrin**, and A. Mastropaolo, “Parameter-efficient multi-task fine-tuning in code-related tasks,” *Under review at ACM Transactions on Software Engineering and Methodology (TOSEM)*, 2026.
- 5 **S. Afrin**, J. Call, K.-N. Nguyen, O. Chaparro, and A. Mastropaolo, “Resource-efficient & effective code summarization,” in *2025 IEEE/ACM Second International Conference on AI Foundation Models and Software Engineering (Forge)*, IEEE, 2025, pp. 224–235.
- 6 **S. Afrin**, M. Z. Haque, and A. Mastropaolo, “A systematic literature review of parameter-efficient fine-tuning for large code models,” *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 2025. DOI: 10.1145/3796522.
- 7 **S. Afrin**, B. Xu, and A. Mastropaolo, “Is quantization a deal-breaker? empirical insights from large code models,” in *2025 IEEE International Conference on Software Maintenance and Evolution (ICSME)*, IEEE, 2025, pp. 1–13. DOI: 10.1109/ICSME64153.2025.00049.
- 8 Y. Gong, A. Tarafder, **S. Afrin**, and P. Kumar, “Single-gpu gnn systems: Traps and pitfalls,” *arXiv preprint arXiv:2402.03548*, 2024.

Research Publications (Prior)

- 1 **S. Afrin**, F. J. M. Shamrat, T. I. Nibir, M. F. Muntasim, M. S. Moharram, M. Imran, and M. Abdulla, “Supervised machine learning based liver disease prediction approach with lasso feature selection,” *Bulletin of Electrical Engineering and Informatics*, vol. 10, no. 6, pp. 3369–3376, 2021.
- 2 S. Chakraborty, F. J. M. Shamrat, **S. Afrin**, S. Saha, I. Ahmed, and S. Thapa, “Industrial fault detection using transfer learning models,” in *2021 2nd International Conference on Smart Electronics and Communication (ICOSEC)*, IEEE, 2021, pp. 1–6.

- 3 P. Ghosh, A. Karim, S. T. Atik, **S. Afrin**, and M. Saifuzzaman, "Expert cancer model using supervised algorithms with a lasso selection approach," *International Journal of Electrical and Computer Engineering*, vol. 11, no. 3, pp. 2632–2640, 2021.
- 4 P. Ghosh, F. J. M. Shamrat, S. Shultana, **S. Afrin**, A. A. Anjum, and A. A. Khan, "Optimization of prediction method of chronic kidney disease using machine learning algorithm," in *2020 15th international joint symposium on artificial intelligence and natural language processing (iSAI-NLP)*, IEEE, 2020, pp. 1–6.

Employment History

- Summer 2026  **Research Intern**, Università del Sannio, Benevento, Italy. Summer research internship supervised by Prof. Massimiliano Di Penta (June – September 2026).
- 2024 – Present  **Research Assistant**, William & Mary.
- 2022 – 2024  **Teaching Assistant**, William & Mary.
- 2019 – 2022  **Lecturer**. Department of Computer Science and Engineering, Daffodil International University, Bangladesh.
- 2018 – 2019  **Project Intern**. Bangladesh Venture Capital LTD,

Skills

- Languages  Strong reading, writing and speaking competencies for English, Bangla.
- Coding  Java, Python, R, C, C++, CUDA, $\LaTeX$
- Misc.  Academic research, teaching, training, consultation, and publishing.

Professional Service

Program Committee Member

- 2026  **PC Member**, Replication and Negative Results (RENE) Track, International Conference on Software Maintenance and Evolution (ICSME) 2026.
-  **PC Member**, Workshop on Agentic Software Engineering (AgenticSE), CAIS 2026.
-  **PC Member**, Mining Challenge Papers Track, Mining Software Repositories (MSR) 2026.
-  **Junior PC Member**, Mining Software Repositories (MSR) 2026.
- 2025  **PC Member**, International Workshop on Analytics for Software Product and Process Improvement (A-SPPI 2025), co-located with PROFES 2025.

Journal Reviewer

-  ACM Transactions on Software Engineering and Methodology (TOSEM)
-  IEEE Transactions on Software Engineering (TSE)
-  Empirical Software Engineering (EMSE), Springer
-  Journal of Systems and Software (JSS), Elsevier

Professional Service

Articles & Columns

- 2026  **Polyglot but Perilous: A Multi-Model, Multi-Language Audit of LLM-Generated Code**, “Spotlight on Transactions” column, *Computer* (IEEE Computer Society), with A. Mastropaolo.
DOI: 10.1109/MC.2026.3696004.

Miscellaneous Experience

Awards and Achievements

- 2026  **Distinguished Junior PC Reviewer Award**, 23rd International Conference on Mining Software Repositories (MSR 2026), co-located with ICSE 2026, Rio de Janeiro, Brazil.
- 2025  **NSF Student Travel Grant**, 41st IEEE International Conference on Software Maintenance and Evolution (ICSME 2025), Auckland, New Zealand.
- 2018  **Dr. Aminul Islam Scholarship**, Selected as the top candidate.
- 2015 – 2019  **Department Scholarship for Outstanding Student Performance**, Daffodil International University.

Seminars and Events

- 2022  **Volunteer, CS Symposium, Department of Computer Science** . William & Mary
- 2023  **Volunteer, CS Symposium, Department of Computer Science** . William & Mary
- 2024  **Volunteer, CS Symposium, Department of Computer Science** . William & Mary
- 2020  **Served as a Rapporteur of a technical session titled Technical Session V-Automation & Robotics** . in 4th International Joint Conference on Advances in Computational Intelligence (IJCACI 2020)
- 2019  **Attended Blockchain Research Seminar**. organized by DIU Blockchain Research and Development Lab.
-  **Performed as a judge** . for the hackathon program at the Daffodil ICT Carnival, 2019.
-  **Participated YES**. Young Entrepreneur Summit, Bangladesh, 2019.

References

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